



AC 299r Special Topics Course Description Form

Juntang Wang

juntangwang@g.harvard.edu

Student Name

Student Email

Chirag Patel

chirag_patel@hms.harvard.edu

Research Advisor Name

Research Advisor Email

Fall



2026

Course Term

Year

Date Submitted

Instructions:

1. Answer the questions below. For both, you may attach additional pages if necessary.
2. Obtain the approval of your Research Advisor, as indicated by a signature below.
3. Return the completed form to the Academic Operations Administrator in the Office of Master's and Professional Programs (SEC 1.312-12) or via email (mastersprograms@seas.harvard.edu).
4. Request permission to enroll in AC 299r in my.harvard. (Permission will be granted after your form has been processed, usually 1-2 business days.)

Be sure to complete the above steps with enough time for your enrollment request to be approved, and for you to return to my.harvard to complete the enrollment process, **by the Course Registration deadline** for the relevant term.

It is suggested that the student and research advisor meet weekly or bi-weekly to discuss progress. Note that AC 299r is a letter-graded course.

TOPICS TO BE COVERED IN THE COURSE

Please provide an outline of the goals of the independent study and the major themes.

Foundation Models and Agentic Systems for Wearable Health Data. Two independently viable projects. AnyWearG: can scarce paired CGM supervision metabolically specialize an existing wearable foundation model while preserving its broad capabilities? StateGlu: does explicit learned physiological state, with outcome-conditioned longitudinal memory, improve a tool-using health agent's reasoning over an otherwise identical stateless agent? Neither project depends on the other. Full description on the attached page 2.

COURSE MATERIALS AND/OR DELIVERABLES

List of texts, reading materials, and/or research articles that will be read or description of experiments or other work that will be performed.

Readings (14 articles on wearable and CGM foundation models, tool-using agents and measurement), the work to be performed for each project, the final product (two conference-format papers with a reproducible codebase), the meeting schedule, a week-by-week timetable for Fall 2026 and a proposed grade basis are all set out on the attached page 2.

9/2/2026

Research Advisor Signature

Date

AC 299r Special Topics Course Description

Student Juntang Wang, juntangwang@g.harvard.edu
Research advisor Chirag Patel, HMS/DBMI
Term Fall 2026

Topics to be covered in the course

Foundation Models and Agentic Systems for Wearable Health Data. Two independently viable projects:

AnyWearG. Can scarce paired CGM supervision metabolically specialize an existing wearable foundation model while preserving its broad capabilities? CGM is privileged supervision during post-training only; wearable-only inference is the intended end state. A preliminary probe found no recoverable glycemic signal in an off-the-shelf PPG foundation model, which is why supervision rather than pretraining is the object of study.

StateGlu. Does explicit learned physiological state, with outcome-conditioned longitudinal memory, improve a tool-using health agent's reasoning about glucose, meals and behavior over an otherwise identical stateless agent?

Neither project depends on the other. The independent-study format is necessary because both involve open-ended research questions and iterative experimentation that existing coursework does not cover.

Course materials and/or deliverables

Readings. *Wearable foundation models:* Abbaspourazad et al., arXiv 2312.05409; Narayanswamy et al., arXiv 2410.13638; Zhang et al., SensorLM, arXiv 2506.09108; Nie et al., AnyPPG, arXiv 2511.01747. *CGM models and forecasting:* Lutsker et al., Nature 2026; Luo et al., npj Health Systems 2025; Isaac, Collin and Patel, SSM-CGM, arXiv 2510.04386. *Agents and their evaluation:* Healey and Kohane, PSB 2025; Deng et al., AlphaQuanter, arXiv 2510.14264. *Measurement and clinical targets:* Patel, Burford and Ioannidis, J Clin Epidemiol 2015; Battelino et al., Diabetes Care 2019; Das et al., Scientific Data 2025.

Work to be performed. *AnyWearG:* baseline probes of pretrained wearable foundation models; a plug-and-play CGM-guided specialization component that aligns the wearable representation to a CGM latent; downstream evaluation; a retention battery on each backbone's original tasks; replication on a second backbone. *StateGlu:* a state-and-memory loop over existing tools; an ablation holding tools and language backbone fixed while varying only state and memory; expert rating on correctness, safety and actionability.

Final product. Two conference-format research papers, 8 to 15 pages each, with a reproducible codebase.

Meetings. Weekly with the research advisor, 30 to 60 minutes. About 15 hours of work per week.

Schedule. Fall 2026, classes begin 2 September.

Weeks	Work
1–2	Literature review; backbone and dataset selection; baseline reproduction
3–4	AnyWearG specialization component; StateGlu tool inventory and state interface
5–6	AnyWearG downstream evaluation and retention battery; Paper 1 draft
7–8	Midterm review; second backbone; StateGlu ablation harness
9–10	AnyWearG replication; StateGlu expert-rating protocol
11–13	Final experiments; both papers written; reproducibility checks

Proposed grade basis.

Weekly meetings and written progress notes 10%; annotated bibliography 10%; implementation and reproducible codebase 20%; midterm research review 15%; Paper 1, AnyWearG 25%; Paper 2, StateGlu 20%.